

Unit 1: Introduction to material science, classification of materials, **Important Properties of engineering materials and materials structure**

Material Science and Engineering

Materials science involves investigating the relationships that exist between the structures and properties of materials.

In contrast, materials engineering is, on the basis of these structure–property correlations, designing or engineering the structure of a material to produce a predetermined set of properties.

From a functional perspective, the role of a materials scientist is to develop or synthesize new materials, whereas a materials engineer is called upon to create new products or systems using existing materials, and/or to develop techniques for processing materials.

Meaning of structure

The structure of a material usually relates to the arrangement of its internal components. Subatomic structure involves electrons within the individual atoms and interactions with their nuclei.

On an atomic level, structure encompasses the organization of atoms or molecules relative to one another.

The next larger structural realm, which contains large groups of atoms that are normally agglomerated together, is termed microscopic, meaning that which is subject to direct observation using some type of microscope.

Finally, structural elements that may be viewed with the naked eye are termed macroscopic.

Meaning of Property

All materials are exposed to external stimuli that evoke some type of response. For example, a specimen subjected to forces will experience deformation, or a polished metal surface will reflect light. A property is a material trait in terms of the kind and magnitude of response to a specific imposed stimulus. Generally, definitions of properties are made independent of material shape and size.

Important properties of solid materials may be grouped into six different categories: mechanical, electrical, thermal, magnetic, optical, and deteriorative.

Mechanical properties relate deformation to an applied load or force; examples include elastic modulus (stiffness), strength, and toughness.

For electrical properties, such as electrical conductivity and dielectric constant, the stimulus is an electric field.

The thermal behavior of solids can be represented in terms of heat capacity and thermal conductivity.

Magnetic properties demonstrate the response of a material to the application of a magnetic field.

For optical properties, the stimulus is electromagnetic or light radiation; index of refraction and reflectivity are representative optical properties.

Finally, deteriorative characteristics relate to the chemical reactivity of materials.

In addition to structure and properties, two other important components are involved in the science and engineering of materials—namely, processing and performance.

The structure of a material will depend on how it is processed. Furthermore, a material's performance will be a function of its properties.



Classification of materials:

Solid materials have been conveniently grouped into three basic categories: metals, ceramics, and polymers. Another category is advanced materials—those used in high-technology applications, such as semiconductors, biomaterials, smart materials, and nanoengineered materials

Metals:

- Materials in this group are composed of one or more metallic elements (e.g., iron, aluminum, copper, titanium, gold, and nickel), and often also nonmetallic elements (e.g., carbon, nitrogen, and oxygen) in relatively small amounts.
- Atoms in metals and their alloys are arranged in a very orderly manner.
- These materials are relatively stiff and strong, yet are ductile and are resistant to fracture which accounts for their widespread use in structural applications.
- Metals are extremely good conductors of electricity and heat, and are not transparent to visible light

Ceramics:

- Ceramics are compounds between metallic and nonmetallic elements; they are most frequently oxides, nitrides, and carbides. Eg. aluminum oxide (or alumina, Al_2O_3), silicon dioxide (or silica, SiO_2), silicon carbide (SiC), silicon nitride (Si_3N_4).
- Traditional ceramics—those composed of clay minerals (i.e., porcelain), as well as cement and glass.
- Ceramic materials are relatively stiff and strong—stiffnesses and strengths are comparable to those of the metals.
- They are typically very hard.
- Historically, ceramics have exhibited extreme brittleness (lack of ductility) and are highly susceptible to fracture. However, newer ceramics are being engineered to have improved resistance to fracture; these materials are used for cookware, cutlery, and even automobile engine parts.
- Ceramic materials are typically insulative to the passage of heat and electricity, and are more resistant to high temperatures and harsh environments.
- Ceramics may be transparent, translucent, or opaque, and some of the oxide ceramics (e.g., Fe_3O_4) exhibit magnetic behavior.

Polymers:

- Polymers include the familiar plastic and rubber materials. Many of them are organic compounds that are chemically based on carbon, hydrogen, and other non-metallic elements (i.e., O, N, and Si).
- They have very large molecular structures, often chainlike in nature, that often have a backbone of carbon atoms.
- Some of the common and familiar polymers are polyethylene (PE), nylon, poly(vinyl chloride) (PVC), polycarbonate (PC), polystyrene (PS), and silicone rubber.
- Typically have low densities.
- Many of the polymers are extremely ductile and pliable (i.e., plastic), which means they are easily formed into complex shapes.
- One major drawback to the polymers is their tendency to soften and/or decompose at modest temperatures, which, in some instances, limits their use.
- They have low electrical conductivities and are nonmagnetic.

Composites:

- A composite is composed of two (or more) individual materials, which come from the categories previously discussed—metals, ceramics, and polymers. The design goal of a composite is to achieve a combination of properties that is not displayed by any single material, and also to incorporate the best characteristics of each of the component materials.
- One of the most common and familiar composites is fiberglass, in which small glass fibers are embedded within a polymeric material (normally an epoxy or polyester). The glass fibers are relatively strong and stiff (but also brittle), whereas the polymer is more flexible. Thus, fiberglass is relatively stiff, and flexible.
- Another technologically important material is the carbon fiber–reinforced polymer (CFRP) composite—carbon fibers that are embedded within a polymer. These materials are stiffer and stronger than glass fiber–reinforced materials, but more expensive. CFRP composites are used in some aircraft and aerospace applications, as well as high-tech sporting equipment (e.g., bicycles, golf clubs, tennis rackets, and skis/snowboards) etc.

Advanced Materials:

Materials that are utilized in high-technology (or high-tech) applications are termed advanced materials. Advanced materials are typically traditional materials whose properties have been enhanced, and also newly developed, high-performance materials. Furthermore, they may be of all material types (e.g., metals, ceramics, polymers), and are normally expensive. Advanced materials include semiconductors, biomaterials, and what we may term “materials of the future”.

a. Semiconductors

- Semiconductors have electrical properties that are intermediate between the electrical conductors (i.e., metals and metal alloys) and insulators (i.e., ceramics and polymers).
- The electrical characteristics of semiconductors are extremely sensitive to the presence of minute concentrations of impurity atoms, for which the concentrations may be controlled over very small spatial regions.

- Semiconductors have made possible the advent of integrated circuitry that has totally revolutionized the electronics and computer industries (not to mention our lives) over the past three decades.

b. Biomaterials

- Biomaterials are employed in components implanted into the human body to replace diseased or damaged body parts.
- These materials must not produce toxic substances and must be compatible with body tissues (i.e., must not cause adverse biological reactions).
- All of the preceding materials—metals, ceramics, polymers, composites, and semiconductors—may be used as biomaterials.

c. Smart Materials

- Smart (or intelligent) materials are a group of new and state-of-the-art materials now being developed that will have a significant influence on many of our technologies.
- The adjective smart implies that these materials are able to sense changes in their environment and then respond to these changes in predetermined manners—traits that are also found in living organisms.
- Components of a smart material (or system) include some type of sensor (that detects an input signal), and an actuator (that performs a responsive and adaptive function).

d. Nanomaterials

- Nanomaterials are materials with at least one dimension in the nanometer scale (1–100 nm), leading to unique physical, chemical, and biological properties not seen in their bulk counterparts.
- Due to their small size and high surface-to-volume ratio, nanomaterials exhibit exceptional properties such as: Increased strength and flexibility (e.g., carbon nanotubes), Enhanced light absorption or emission (e.g., quantum dots), Improved conductivity or semiconducting behavior (e.g., graphene).
- Types:
 - Carbon-Based: Fullerenes, graphene, carbon nanotubes.
 - Metal-Based: Gold and silver nanoparticles, metal oxides (e.g., TiO₂, ZnO).
 - Dendrimers: Branched organic molecules used in drug delivery.
 - Composites: Combinations of nanomaterials to enhance properties.
- Applications:
 - Electronics: High-performance transistors, memory devices, and sensors
 - Energy: Solar cells, batteries, and supercapacitors.
 - Healthcare: Drug delivery, imaging, and diagnostic tools.
 - Environment: Water purification and pollution control.
- Fabrication Methods:
 - Top-Down Approach: Reducing bulk materials to the nanoscale (e.g., lithography).
 - Bottom-Up Approach: Building structures atom by atom or molecule by molecule (e.g., chemical vapor deposition).